TrustEpisode: Deterministic and Auditable Reconstruction of EDR/NDR Attack Episodes

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Abstract

Investigators reconstructing multi-step attacks from EDR and NDR telemetry need a deterministic, auditable reconstruction contract: single-owner episode membership, immutable revisions under late evidence, collector health distinct from "no detection," and a replayable terminal outcome per revision.

We present TrustEpisode for sealed EDR/NDR attack-episode reconstruction. On the sealed primary live CALDERA cohort (60 M1–M6 malicious runs; 40 standalone benign), sliding sessionization raises mean coverage from 0.942 to 0.958 versus fixed 300 s bins, and fair sliding baselines EB1–EB4 show *no observed difference* (60/60 paired ties; not an equivalence test). Canonical ordered replay is byte-identical on all 60 sealed malicious runs (60/60); raw delivery order is not order-robust without re-canonicalization. Audit conformance covers all seven AssemblyFailureRecord codes (18/18 fixtures). The A–G evidence-loss taxonomy is conformance-tested (8/8) and instantiated on sealed M2 as D_token_mapping_failure. Contract-ablation baselines AB0–AB3 isolate which properties prevent history mutation, silent failure, and noncanonical replay. A separate supplemental partition reports live L1/L4 bases, offline L2/L3 ablations (including controlled hub load), controlled late inject, and wall-clock forwarder holds (within-horizon 2/2; beyond-horizon sealed but LateEvidenceRecord not observed organically). Sealed M7 is reported only as a supplemental 3900 s boundary-stress cohort (coverage 0 under W_{max} = 3600 s), not as a primary estimator. Fitted scoring, calibration, and outage robustness remain out of scope.

Keywords: digital forensics, attack episode reconstruction, EDR, NDR, replayable audit, evidence provenance

1. Introduction

Security operations and digital-forensic investigations reconstruct multi-step attacks from temporally dispersed endpoint and network observations [13, 9, 1]. Isolated EDR or NDR records may appear benign alone yet become material when linked by time, entity, session, and process relations. Reconstruction is not only a clustering problem: investigators need a contract that fixes who owns episode membership, how late evidence is admitted, how collector health differs from "no detection," and how every revision leaves an independently verifiable terminal outcome [4, 3, 14].

Prior alert-correlation and provenance systems motivate multi-stage context using rules, graphs, and learned representations [10, 8, 6, 7, 5, 17]. Those lines do not remove the need for fixed ownership of membership, immutable revision history, and replayable audit semantics under incomplete observability. This submission evaluates TrustEpisode as a *deterministic, auditable reconstruction contract* for EDR/NDR evidence. A compact scorer and calibrator remain optional schema interfaces and are not empirically validated here.

Research questions..

- **RQ1** Under identical sealed EDR/NDR evidence, does TrustEpisode reproduce byte-identical membership, revision history, late-evidence records, and digests on all 60 malicious runs?
- **RQ2** Do open, finalized, late-successor, closed, beyond-horizon, duplicate, and all seven AssemblyFailureRecord codes yield explicit immutable audit outcomes without rewriting earlier history?
- **RQ3** When a required action is absent from a reconstructed episode, can the pipeline localize loss across the closed A–G taxonomy, and is that taxonomy instantiated on sealed live evidence?
- **RQ4** Which relation, hub, merge, and late-revision mechanisms does the live M1–M6 cohort exercise, and which require targeted conformance cards?
- **RQ5** What portion of observed formation gain comes from replacing fixed bins with sliding sessionization, and does relation-aware logic provide additional measurable benefit on the sealed cohort?

Contributions..

1. a single-owner immutable reconstruction contract: Episode Synthesis alone owns membership; late evidence yields successors or LateEvidenceRecords; deadlines do not extend; terminal history is append-only;
2. replayable forensic audit outcomes with exactly one terminal AssemblyOutcome per revision, canonical references, SHA-256 digests, and a complete seven-code failure matrix;
3. a stage-localized A–G evidence-loss taxonomy, conformance-tested across classes and instantiated on sealed M2 as `D_token_mapping_failure`;

4. a fair negative-result evaluation: legacy fixed bins underperform sliding sessionization, while fair sliding EB1–EB4 show no observed difference on 60 sealed runs; advanced guards require targeted cards not mixed into primary effectiveness;
5. a determinism boundary on all 60 sealed malicious runs: canonical ordered replay and duplicate retries are stable; raw delivery-order permutations are not order-robust without re-canonicalization; formation is single-writer per run;
6. a contract-ablation comparison (AB0–AB3) isolating which forensic failure modes mutable, noncanonical, and silent-failure stores admit relative to TrustEpisode.

2. Related Work

2.1. Alert Correlation and Attack Reconstruction

Classical alert correlation reconstructs multi-step scenarios from intrusion alerts by temporal and causal linking [10]. Provenance-based detectors such as HOLMES, UNICORN, KAIROS, NODLINK, and CAPTAIN recover APT context from system graphs and alerts [8, 6, 5, 7, 17]. Those systems motivate episode construction; they do not by themselves fix single-owner membership, late-data revision policy, or append-only terminal audit outcomes for mixed EDR/NDR streams.

2.2. Digital Forensic Event Reconstruction

Timeline-based event reconstruction is a core digital-forensic technique for inferring past activity from heterogeneous artifacts [1, 4]. Recent work formalizes timestamp interpretation via time anchors and clock-skew anomalies [14] and surveys challenges beyond mere data volume [1]. Process models and filesystem analysis emphasize documenting what was examined, what was unavailable, and how conclusions depend on incomplete traces [3, 9]. TrustEpisode imports these audit requirements into *online* EDR/NDR episode reconstruction: immutable revisions, explicit late stores, and digestable terminal outcomes under delayed delivery.

2.3. Reproducibility, Tool Validation, and Provenance

Forensic tool and method validation requires documented procedures, known datasets, and repeatable results before casework [12, 2, 16]. Chain-of-custody and provenance discipline demand that investigators show integrity of examined objects and distinguish observed from unavailable evidence [4, 9]. TrustEpisode's contribution is not a higher detector F1; it is a single-owner revision contract, canonical replay boundary, and explicit terminal failure semantics that make reconstruction outcomes auditable under the same reproducibility expectations.

2.4. Incomplete Telemetry and Evidence-Loss Attribution

EDR and NDR fail differently; missing modality, parser blind spots, and collector health can erase or distort attack steps without implying that correlation "missed" an already-normalized event [19, 15]. Reconstruction contracts must therefore separate sensor observability, normalization/token mapping, formation membership, and offline attribution when diagnosing incomplete episodes.

2.5. Positioning Summary

Table 1 contrasts representative lines along forensic-contract properties. Unknown cells are marked *Not reported*; we do not infer undocumented product behavior. TrustEpisode's novelty is the audit lifecycle and evidence-loss localization contract, not a new graph algorithm or classifier.

2.6. Scope Distinction

TrustEpisode does not claim a new graph detector, AI ranking effectiveness, attribution beyond observable EDR/NDR, enterprise impact estimation, or autonomous response. The evaluated contribution is a deterministic forensic reconstruction and audit contract for sealed EDR/NDR evidence under late arrival and incomplete observability.

3. TrustEpisode Reconstruction Contract

TrustEpisode converts EDR/NDR telemetry into deterministic episode revisions and version-scoped forensic audit records. Evidence Fabric and Health Monitor own events and source health; Episode Synthesis alone owns membership; optional Feature Extractor / Scorer / Calibrator fields remain schema interfaces without fitted artifacts in this submission. For each revision, the AuditRecord Assembler emits exactly one terminal AssemblyOutcome: a canonical AuditRecord or an immutable AssemblyFailureRecord. Ground truth is joined only offline. Figure 1 shows these boundaries.

3.1. Investigation and Threat Model

Assumptions.. Raw pointers and the sealed evidence store are integrity-checkable; Health Monitor coverage messages have authenticated sources; attackers may induce missing events, delayed delivery, and duplicate retries; attackers cannot arbitrarily forge signed health state; runtime software may fault; the offline evaluator is isolated from online formation.

Non-assumptions.. We do *not* assume complete telemetry, that collector silence equals downtime, that all attacks are observable, that ground truth is available online, that model scores are reliable, or that raw arrival order is fixed.

Investigator workflow.. (1) import sealed EDR/NDR; (2) form EpisodeRevisions; (3) inspect the terminal audit outcome; (4) check coverage and provenance; (5) accept a late successor or LateEvidenceRecord; (6) if a required step is missing, run the A–G waterfall; (7) replay and verify digests; (8) report observed, unavailable, and unknown separately.

3.2. Evidence and Reconstruction Model

Online objects are NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, AuditRecord, and AssemblyFailureRecord. Offline labels never enter online schemas. Table 2 assigns every object to one producer.

Table 1: Related-work comparison on forensic reconstruction contract properties (Not reported = not documented in the cited source).

Work	EDR+NDR	Imm. rev.	Late evid.	Canon. replay	Term. fail	Src health	Loss loc.	Live
Alert correlation [10]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
HOLMES [8]	Partial	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
UNICORN [6]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
KAIROS [5]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
Timeline SoK [1]	Survey	Survey	Survey	Survey	Survey	Survey	Survey	Survey
Time anchors [14]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Clock skew	Yes
TrustEpisode (this)	Yes	Yes	Yes	Yes	Yes	Yes	A–G	Yes

Table 2: Single-owner online object contract.

Object	Sole producer	Consumers	Forbidden behavior
NormalizedEvent	Evidence Fabric	Episode Synthesis	label, scenario, or health fields
SourceCoverage	Health Monitor	Scope Resolver; Sufficiency; Assembler	infer outage from event silence
EpisodeRevision / LateEvidenceRecord	Episode Synthesis	Feature; Sufficiency; Assembler / late store	rewrite a prior revision; admit beyond-horizon into membership
FeatureVector $[D, C, m]$	Feature Extractor	Scorer; Assembler	labels or inferred health
Nullable P	Calibrator (design)	Assembler	numeric unsupported output
Structured U	Sufficiency Evaluator	Assembler	scalarization that rewrites P
AssemblyOutcome	AuditRecord Assembler	Append-only store; replay	both success and failure; update/delete

EDR/NDR events + authenticated source health

↓

Evidence Fabric / Health Monitor

↓

Episode Synthesis (sole membership owner)

↓

AuditRecord Assembler → append-only store

Offline only: labels, matching, evaluation.

Figure 1: Label-blind online reconstruction and offline evaluation boundary.

3.3. Independent Evidence Health and Observability

Event silence is not an outage. "No detection" is not "unavailable." SourceCoverage carries authenticated health, parser validity, and an observed source mask. Assemblers consume exact coverage references; they do not infer health from empty event windows.

3.4. Deterministic Episode Formation

Formation links events using process-causal, authenticated-session, endpoint-connection, and specific-entity relations under a 300 s maximum inter-event gap and 3600 s maximum episode span. Tie-breaks are deterministic (relation precedence, relation count, Δt, start time, episode ID). Hub suppression, parent merge, and exact event-time ordering are part of the frozen policy. Formation never reads labels, receipts, or scenario metadata.

3.5. Immutable Revision Lifecycle

Watermark lag finalizes open episodes. Late evidence inside a fixed 900 s revision horizon may create an immutable successor revision; beyond-horizon or closed-lineage evidence produces a LateEvidenceRecord. Earlier revision digests are never rewritten. Deadlines are not extended by late arrivals.

3.6. Terminal Forensic Audit Contract

At `revision.watermark+120` s the assembler inserts exactly one terminal outcome under ($episode\_id, revision$): AuditRecord or AssemblyFailureRecord with a closed failure code set (schema, key, duplicate, digest, version, missing-input, timeout). Success cannot follow terminal failure. References are canonically ordered; RFC 8785/JCS serialization and SHA-256 digests bind versions. The store is append-only.

Table 3: Evidence-loss taxonomy for incomplete reconstruction.

Class	Decision rule (summary)
`A_execution_failure`	Action receipt/predicate shows non-success
`B_sensor_observability_failure`	Receipt success but raw EDR/NDR lacks identifiable activity
`C_parser_or_normalization_failure`	Raw present; NormalizedEvent absent or rejected
`D_token_mapping_failure`	NormalizedEvent present under wrong/missing required token
`E_formation_membership_failure`	Correct token exists but not retained in matched episode
`F_offline_attribution_failure`	Episode retains evidence; matcher fails to credit token
`G_unknown_or_incomplete_trace`	Required supporting records missing for A–F

3.7. Evidence-Loss Taxonomy

When a required action token is absent from a reconstructed episode, diagnosis uses the following closed classes (Table 3):

Forbidden inference: treating F1 = 1.0 as exact-complete step recovery; treating missing tokens as formation failures without a waterfall.

3.8. Scoring and Probability Extension (Design Only)

Schema fields may retain an optional score interface: version-bound, nullable P outside support, and U that must not rewrite P. This submission reports no fitted coefficients, Brier/NLL, or held-out ranking. Scoring and calibration are not empirical claims here.

3.9. Boundary

The *evaluated* contribution is deterministic episode reconstruction and replayable audit semantics, including evidence-loss localization. Manifest-scoped probability is design-only. CTI, RAG/agents, impact scoring, and OT/ICS generalization are excluded.

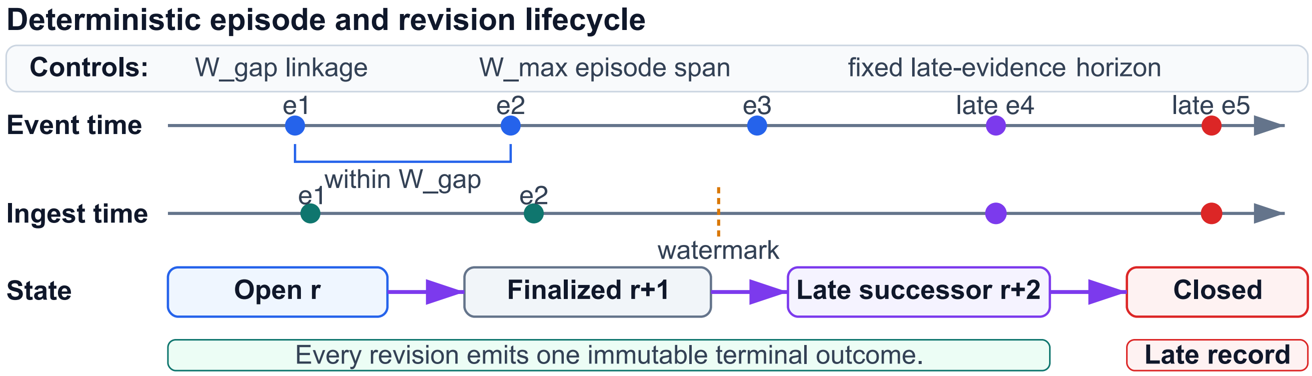

Figure 2: Episode revision lifecycle: open → finalized → late successor or closed; beyond-horizon evidence enters LateEvidenceRecord without rewriting earlier membership.

4. Evaluation

4.1. Evidence Classes and Cohorts

- **RQ1** reproducibility — sealed offline replay on all 60 primary malicious runs.
- **RQ2** lifecycle auditability — A1–A12 synthetic audit conformance covering seven failure codes + late/beyond-horizon cards.
- **RQ3** evidence-loss localization — A–G taxonomy conformance + sealed M2 waterfall.
- **RQ4** mechanism coverage — live M1–M6 + synthetic T1–T8; M7 only as supplemental boundary stress.
- **RQ5** baseline interpretation — sealed fair EB0–EB4 comparison on M1–M6.

Evidence classes used below are disjoint: (i) primary live empirical (M1–M6 formation cells); (ii) sealed offline replay on those primary seals; (iii) synthetic conformance (A1–A12, T1–T8, AB0–AB3, A–G fixtures); (iv) unit/integration boundary fixtures; (v) supplemental partition artifacts (L1–L4, wall-clock, sealed M7) that are *not* mixed into primary estimators. Synthetic cards are never mixed into primary live effectiveness. Sealed Table 5 primary cells were not overwritten.

4.2. Primary Live M1–M6 Cohort

CALDERA v5.3.0 executes frozen Bash abilities on two isolated Ubuntu 22.04 targets with an EDR wrapper and inline Suricata NDR. Online schemas reject labels and receipts. The primary sealed cohort has 60 malicious runs (M1–M6 × two modes × five) and 40 standalone benign runs. Fair sliding baselines and paired comparisons are computed on this M1–M6 set only.

4.3. Ground Truth and Matching

Offline matching uses temporal IoU ≥ 0.10, entity Jaccard ≥ 0.50, overlapping required action tokens, and weight

$$w = 0.5A_{cov} + 0.3J_{entity} + 0.2\,\mathrm{IoU}_{time}; \quad w \geq \theta_{match} = 0.50. \quad (1)$$

Exact-complete recovery requires $A_{cov} = 1$. Online builders never read evaluation fields.

4.4. Fair Reconstruction Baselines

- **EB0_FIXED_BIN_300s** (legacy): fixed bins; boundary reference only.
- **EB1_SLIDING_TIME**: same $W_{gap} = 300$ s, $W_{max} = 3600$ s; time only.
- **EB2_SLIDING_ENTITY**: same lifecycle; entity overlap only.
- **EB3_RELATION_NO_ADVANCED_GUARDS**: relations on; hub/merge/late off.
- **EB4_TRUSTEPISODE**: full production policy.

Primary comparison is EB1–EB4 on identical sealed streams, aliases, matching, and ground truth. Configuration digests are sealed in `results/derived/fair_baselines.json`.

4.5. Primary Formation Results and Fair Negative Result

Table 5 reports production-formation cells for the primary live M1–M6 cohort ($n = 5$ per cell). Concurrent-benign mean contamination remains 0.356 for fair builders.

Relative to legacy EB0, sliding policies raise coverage (0.942→0.958) and exact-complete recovery (0.783→0.833) on the sealed primary cohort. Fixed bins cut multi-minute chains at arbitrary boundaries; sliding gap/sessionization preserves those chains under the same span cap.

Under fair sliding baselines, EB1–EB4 are identical on every primary metric and every family: paired EB4–EB1 differences are zero with 60 ties and CIs $[0, 0]$. We report *no observed difference* (not a formal equivalence test). Relation-aware formation is *not* reported as better than fair sliding-time.

Why sliding-time already suffices on M1–M6.. The sealed primary cohort does not make advanced relations the unique linkage path: time and entity sessionization already recover the same matched episodes. Hub, parent-merge, and late guards therefore show no end-to-end metric delta. End-to-end F1 alone can misleadingly suggest that complex policy is “validated” when the benchmark never stresses it. Future BAS cohorts should register mechanism-coverage requirements (session-only, endpoint-only, hub over-merge, late horizon, beyond-W_{max} span) before claiming relation-aware gains.

Table 4: Primary M1–M6 scenario families (simplified). "Live EB1–EB4 delta?" reports whether fair builders differed on the sealed primary cohort for the intended advanced mechanism.

ID	Ordered activity (summary)	Expected EDR/NDR	Intended mechanism	Live EB1–EB4?
M1	recon → local action	process; DNS/flow	entity/time	identical
M2	remote session + SCP archive	process; SSH/SCP flows	session / file_transfer token	identical; token gap
M3	privilege / persistence steps	process; file	causal/entity	identical
M4	lateral-style dual-host steps	process; flows	session/endpoint	identical
M5	staged tooling / execution	process; file/net	entity/time	identical
M6	cleanup / cover tracks	process; file	entity/time	identical

Table 5: Primary live M1–M6 cells under production formation (mean ± sample SD; $n = 5$).

Family	Mode	Coverage	Frag.	Merge	Contam.	F1
M1	attack-only	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
M1	concurrent-benign	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.300 ± 0.112	1.000 ± 0.000
M2	attack-only	0.750 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
M2	concurrent-benign	0.750 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.350 ± 0.091	1.000 ± 0.000
M3	attack-only	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
M3	concurrent-benign	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.350 ± 0.091	1.000 ± 0.000
M4	attack-only	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
M4	concurrent-benign	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.367 ± 0.217	1.000 ± 0.000
M5	attack-only	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
M5	concurrent-benign	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.433 ± 0.091	1.000 ± 0.000
M6	attack-only	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	1.000 ± 0.000
M6	concurrent-benign	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.333 ± 0.257	1.000 ± 0.000

Table 6: Fair sliding-gap baselines versus legacy fixed bins on the sealed 60-run M1–M6 malicious cohort (mean ± sample SD). Primary comparisons share $W_{gap} = 300$ s and $W_{max} = 3600$ s.

Builder	Role	Coverage	Exact-comp.	Frag.	Contam.	F1
EB0_FIXED_BIN_300s	legacy	0.942 ± 0.113	0.783 ± 0.415	0.083 ± 0.279	0.178 ± 0.208	0.950 ± 0.120
EB1_SLIDING_TIME	primary	0.958 ± 0.094	0.833 ± 0.376	0.000 ± 0.000	0.178 ± 0.208	1.000 ± 0.000
EB2_SLIDING_ENTITY	primary	0.958 ± 0.094	0.833 ± 0.376	0.000 ± 0.000	0.178 ± 0.208	1.000 ± 0.000
EB3_RELATION_NO_ADVANCED_GUARDS	primary	0.958 ± 0.094	0.833 ± 0.376	0.000 ± 0.000	0.178 ± 0.208	1.000 ± 0.000
EB4_TRUSTEPISODE	primary	0.958 ± 0.094	0.833 ± 0.376	0.000 ± 0.000	0.178 ± 0.208	1.000 ± 0.000

Table 7: Paired differences EB4−EB1_SLIDING_TIME (contamination/fragmentation flipped so positive favors EB4). Bootstrap: 2000 run-cluster resamples, seed 20260721. Zero deltas reported as no observed difference.

Metric	Mean Δ	Median Δ	95% CI	W/T/L	Interpretation
action_coverage	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference
exact_complete_recovery	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference
fragmentation	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference
contamination	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference
episode_f1	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference

Table 8: Inclusive W_{gap}/W_{max} boundary fixtures (unit; all pass).

Dimension	Seconds	Expected	Observed
W_{gap}	299	eligible	eligible
W_{gap}	300	eligible	eligible
W_{gap}	301	ineligible	ineligible
W_{max}	3599	eligible	eligible
W_{max}	3600	eligible	eligible
W_{max}	3601	ineligible	ineligible
W_{max}	3900	ineligible	ineligible

4.6. Inclusive Temporal Boundary Fixtures

Unit fixtures on production `EpisodeSynthesizer._candidate` confirm inclusive equality at the frozen contracts: $W_{gap} \in \{299, 300\}$ eligible and 301 ineligible; $W_{max} \in \{3599, 3600\}$ eligible and $\{3601, 3900\}$ ineligible (Table 8). The 3900 s case matches the sealed M7 schedule offset and is *not* a substitute for live M7 formation metrics. Results are sealed in `results/derived/temporal_boundary_conformance.json`.

4.7. Sixty-Run Determinism Matrix

On all 60 sealed malicious runs (M1–M6 × attack-only/concurrent-benign × five), D1 canonical ordered replay (10× per run; 600 executions) is byte-identical (60/60). D2 separates raw delivery-order robustness (0/60) from re-canonicalized equality (60/60): we do *not* claim raw-order robustness. D3 ingest-delay grids (including inside, at, and beyond the 900 s revision horizon) pass 60/60; D4 duplicate/idempotent/conflict cases pass 60/60. D5 documents single-writer-per-run ownership: multi-run parallelism is not same-run concurrent mutation, and we claim neither lock-free nor multi-writer determinism.

4.8. Audit Failure Conformance (A1–A12)

RQ2 is supported by the full failure matrix: all seven AssemblyFailureRecord codes (7/7) appear with expected terminal outcomes, plus precedence, exclusivity, idempotent success retry, and append-only rejection (18/18 fixtures pass). This is synthetic forensic audit conformance, not field effectiveness.

Table 9: Determinism matrix on all 60 sealed malicious runs (D1–D5).

Test	Runs	Variants	Expected invariant	Pass	Fail	Interpretation
D1 canonical replay	60	10×/run	byte-identical digests	60	0	canonical ordered input
D2 raw delivery order	60	6 perms	equals baseline without reorder	0	60	not order-robust
D2 re-canonicalized	60	6 perms	equals baseline after sort	60	0	canonicalization required
D3 ingest delay	60	horizon grid	successor/late invariants	60	0	prior digest preserved
D4 duplicates	60	conflict/idempotent	reject or no-op	60	0	fail-closed conflicts
D5 single-writer	architecture	ownership	no same-run multi-writer	documented	0	not lock-free concurrency

Table 10: A1–A12 audit conformance fixtures (synthetic; covers all seven AssemblyFailureRecord codes).

Fixture	Expected	Observed	Term.	Prior chg?	Result
A1_SUCCESS	success	success	1	no	pass
A2_SCHEMA_INVALID	schema_invalid	schema_invalid	1	no	pass
A3_KEY_MISMATCH	key_mismatch	key_mismatch	1	no	pass
A4_CONFLICTING_DUPLICATE	conflicting_duplicate	conflicting_duplicate	1	no	pass
A5_DIGEST_MISMATCH	digest_mismatch	digest_mismatch	1	no	pass
A6_VERSION_MISMATCH	version_mismatch	version_mismatch	1	no	pass
A7_MISSING_INPUT	missing_input	missing_input	1	no	pass
A8_ASSEMBLY_TIMEOUT	assembly_timeout	assembly_timeout	1	no	pass
A9_FAILURE_PRECEDENCE_SCHEMA	schema_invalid	schema_invalid	1	no	pass
A9_FAILURE_PRECEDENCE_KEY	key_mismatch	key_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_DUPLICATE	conflicting_duplicate	conflicting_duplicate	1	no	pass
A9_FAILURE_PRECEDENCE_DIGEST	digest_mismatch	digest_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_VERSION	version_mismatch	version_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_MISSING	missing_input	missing_input	1	no	pass
A9_FAILURE_PRECEDENCE_TIMEOUT	assembly_timeout	assembly_timeout	1	no	pass
A10_TERMINAL_EXCLUSIVITY	rejected	rejected	1	no	pass
A11_IDEMPOTENT_SUCCESS_RETRY	idempotent	idempotent	1	no	pass
A12_APPEND_ONLY_STORE	both_rejected	both_rejected	1	no	pass

4.9. Evidence-Loss Taxonomy Validation

The taxonomy is conformance-tested across A–G and instantiated on one sealed live case, M2 (8/8 fixture rows; 1 live). Synthetic passes do not imply field-validated coverage for every class.

4.10. M2 Evidence Waterfall

Final class: D_token_mapping_failure. Action coverage 0.75 with episode F1 1.0 is therefore not evidence that Episode Synthesis dropped an already-normalized `file_transfer` event. Primary sealed results are unchanged; any mapper fix requires a new version digest and a post-analysis cohort.

4.11. Contract Ablation Baselines (AB0–AB3)

AB0–AB3 are reference implementations for an isolation study: which contract property prevents which forensic failure mode. They are reference implementations for isolation, not product comparisons.

4.12. Mechanism Conformance Cards

T1–T8 are synthetic. T2/T3 were redesigned so that `specific_entity` is disabled and session/endpoint becomes the sole legal relation; ablation then fragments. Passes are conformance, not live effectiveness. Live EB1–EB4 remain tied.

4.13. Supplemental Partition (Not Primary)

Artifacts below live under a separate `supplemental_mechanism` partition (or sealed M7 membership). They do *not* enter Table 5 or the 60-run fair baseline estimator. Evidence classes are stated per row.

- **L1 (live base + offline ablation):** three sealed live M3 SSH cross-host bases; offline ablation links under `authenticated_session` alone and fragments when that relation is disabled (3/3). Not a claim of organic session-only BAS stress.
- **L2 (offline ablation on L1 seals):** on the same live SSH seals, `endpoint_connection` alone (entity/session/causal disabled) links denser than ablating endpoint (3/3). Offline-only mechanism check.
- **L3 (offline ablation + controlled inject):** organic hub load is *not* observed (hub-on vs hub-off identical, 0/3 organic). Controlled hub-load inject on the live sealed bases (cutoff+1 peers) makes hub-on vs hub-off diverge (3/3). Controlled inject ≠ organic BAS hub stress.
- **L4 late evidence (two distinct classes):** (i) *controlled late inject* on live seals yields successors and `beyond_revision_horizon` LateEvidenceRecords (3/3); (ii) *wall-clock forwarder hold/release* with real delays: within-horizon 180 s holds seal and show multi-revision successors (2/2 pass); beyond-horizon attempts (930 s hold-release; 1080 s quiet wait) *seal* but do *not* emit LateEvidenceRecords under organic telemetry (2 sealed, 0/2 LateEvidence; new membership formed instead). Beyond-horizon localization is therefore supported by controlled inject, not by wall-clock beyond runs in this lab.
- **M7 supplemental boundary-stress cohort:** ten sealed live runs (5 attack-only + 5 concurrent-benign) with ability offsets 0/300/3900 s. Production formation yields coverage 0, F1 0, and undefined contamination under W_{max} = 3600 s. M7 is a descriptive long-horizon stress cell and is excluded from primary cell/mode aggregates.

Table 11: A–G evidence-loss taxonomy conformance (synthetic A–G plus one sealed live M2 as D).

True class	Predicted	Source	Result
A_execution_failure	A_execution_failure	synthetic	pass
B_sensor_observability_failure	B_sensor_observability_failure	synthetic	pass
C_parser_or_normalization_failure	C_parser_or_normalization_failure	synthetic	pass
D_token_mapping_failure	D_token_mapping_failure	synthetic	pass
E_formation_membership_failure	E_formation_membership_failure	synthetic	pass
F_offline_attribution_failure	F_offline_attribution_failure	synthetic	pass
G_unknown_or_incomplete_trace	G_unknown_or_incomplete_trace	synthetic	pass
D_token_mapping_failure	D_token_mapping_failure	live M2	pass

Table 12: M2 evidence-loss waterfall (10 sealed runs).

Stage	Expected	Observed	Status
CALDERA receipt (`linux.scp_archive`)	10	10	retained
Raw EDR/NDR observability (scp mentions)	10	10	retained
NormalizedEvent as `file_transfer`	10	0	drop
Normalized as `remote_session` instead	10	10	remapped
Episode retains `file_transfer`	10	0	absent token
Episode retains `remote_session`	10	10	retained

4.14. *Artifact Reproducibility*

The supplementary archive rebuilds macros from sealed JSON, regenerates tables, and re-runs D1–D5, A1–A12, A–G, AB0–AB3, T1–T8, temporal boundary fixtures, and `verify_claims.py`. A clean-room extract of the companion ZIP reproduces those checks under `CLEANROOM_REPRODUCTION_REPORT_20260721.md`. Digests are refreshed with each packing; stale hashes are not reused.

4.15. *Limitations*

Two Ubuntu targets; non-commercial EDR; $n = 5$ per cell; controlled benign workloads; no production base rates; fair baselines tied on primary M1–M6; synthetic cards ≠ field validation; raw order not robust; no fitted scorer/calibration; no partial-telemetry/outage; M2 mapping not corrected in the primary cohort. Organic hub stress remains unobserved on live streams. Wall-clock *within*-horizon holds are executed (2/2); wall-clock *beyond*-horizon LateEvidenceRecord under organic telemetry was attempted (2 sealed) but not observed. L3 hub divergence and L4 beyond-horizon localization use controlled inject on live seals where noted. M7 is supplemental only.

Analysis status.. Complete (primary): live M1–M6 reconstruction (100 sealed train runs with B1–B8); fair sliding baselines with paired CI; T1–T8; M2 waterfall; 60-run determinism D1–D5; A1–A12; A–G; AB0–AB3; inclusive boundary fixtures including 3900 s. Complete (supplemental, not primary): L1 live bases + offline L1/L2; L3 controlled hub inject; L4 controlled late inject; wall-clock within 2/2 and beyond 2 sealed/0 LateEvidence; sealed M7 5+5 boundary-stress. Not executed: organic hub-stressed BAS traffic; fitted scorer / RP / PO1.

5. Conclusion

TrustEpisode provides a deterministic, auditable EDR/NDR episode-reconstruction contract: single-owner immutable revisions, append-only terminal outcomes with a complete failure-code matrix, and stage-localized A–G evidence-loss diagnosis. On the sealed M1–M6 cohort, sliding sessionization improves completeness relative to fixed bins, but fair sliding time/entity/relation builders show *no observed difference* from full TrustEpisode (60/60 ties). That negative result is forensic evidence about benchmark coverage: advanced guards require targeted conformance cards, which we provide without treating them as live effectiveness. Canonical replay is byte-identical on all 60 sealed malicious runs; raw arrival-order robustness requires re-canonicalization. M2's missing `file_transfer` token is `D_token_mapping_failure`, not a demonstrated formation drop. Contract ablation shows which properties prevent mutable history, silent missing-input failure, and noncanonical replay. Supplemental partition artifacts (not primary estimators) include L1 live bases with offline L1/L2 ablations, L3 controlled hub inject, L4 controlled late inject, wall-clock within-horizon holds (2/2), and sealed beyond-horizon attempts without organic LateEvidenceRecord. Sealed M7 is a supplemental 3900 s boundary-stress cohort only. Fitted scoring, calibration, and telemetry-outage robustness remain unevaluated and outside these claims.

Generative AI use disclosure.. OpenAI Codex and Cursor agents assisted with language editing, experiment orchestration, consistency checking, and claim-to-evidence audits. The authors independently designed the study, verified code and reported results, and take responsibility for the manuscript.

Appendix A. Frozen Submission Contracts

Appendix A.1. Machine Schema and Audit Assembly

The normative schema-ready in-PDF contract follows JSON Schema Draft 2020-12 closed-object semantics [18] for NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, FeatureVector, SupportDecision, AuditRecord, and AssemblyFailureRecord. FeatureVector owns D, C, m, o^{det} without health input. SupportDecision owns resolved a^{det}, κ, g, detector count n, eligibility reason, `decision_eligible`, EDR/NDR `coverage_ref_ids`, and `health_snapshot_digest`. Count $n = 0$ forces out-of-support, false eligibility, and null P even if both detectors report `no_detection`. AuditRecord requires those outputs, z, P, U, the revision digest, health snapshot, ordered references, exact versions, and canonical digest; out-of-support requires JSON null P and false eligibility. For each revision key the Assembler atomically inserts one terminal outcome: AuditRecord after a complete exact-version join, or AssemblyFailureRecord after conflict or deadline. The assembly deadline is `revision.watermark` + 120 s. The unique-key store is an insert/select sink and cannot later replace failure with success.

Identifiers are ASCII and sort by unsigned UTF-8 bytes. AuditRecord uses string IDs/digests, nonnegative integer revision,

Table 13: Contract ablation / reference-implementation comparison (synthetic isolation study; not SOTA product baselines).

Property	AB0 Mutable	AB1 Noncan.	AB2 Silent	TrustEpisode	Evidence
prior_history_immutable_under_late	no	no	yes	yes	synthetic conformance
byte_identical_replay_under_ref_reorder	yes	no	yes	yes	synthetic conformance
explicit_terminal_failure_on_missing_input	no	no	no	yes	synthetic conformance
append_only_terminal_store	no	yes	yes	yes	synthetic conformance

Table 14: Targeted synthetic mechanism cards (conformance only; not mixed into primary effectiveness).

Card	Result	Expected	Observed
T1-PROCESS-CAUSAL	pass	fragment without causal	full-eps:1, abl-eps:2
T2-AUTH-SESSION	pass	session only links ablation fragm...	session-only:1, abl:2
T3-ENDPOINT-MAPPING	pass	endpoint only links ablation frag...	endpoint-only:1, abl:2
T4-HUB-GUARD	pass	hub guard changes membership or e...	full-eps:4, no-hub-eps:1
T5-PARENT-MERGE	pass	merge enabled creates parent epis...	full-parents:3, abl-parents:0
T6-LATE-SUCCESSOR	pass	disabling late successor increase...	full-rev:4, abl-rev:3, full-late:...
T7-BEYOND-HORIZON	pass	beyond horizon emits LateEvidence...	late:1, reasons:(beyond-revision-...
T8-TIE-BREAK	pass	canonical arrival order yields id...	equal:True; raw-reverse-differs:True

finite $D, P \in [0,1]$, $C \in \{0,1\}$, numeric z, Boolean eligibility, closed enums/maps for κ, U, m, h, a^{det}, ordered reference arrays, and a closed version object. Only P and group key are nullable: out-of-support requires P = JSON null; only supported-group requires a non-null key. NaN, infinities, negative zero, and binary64 overflow are invalid. AssemblyFailureRecord contains no D, C, z, P, U and has exactly one of the seven failure codes defined by the assembly contract (schema, key, duplicate, digest, version, missing-input, timeout). Canonical bytes follow RFC 8785 JSON canonicalization [11]. JCS sorts object properties but preserves array order, so the application sorts every reference array first. SHA-256 includes each outcome's declared digest-field set except `canonical_digest`. AuditRecord includes type/schema/key/revision, revision digest, $D, C, z, P, \kappa, g, e, U, m, h, o^{det}, a^{det}$, detector count/reason, ordered references, and versions; AssemblyFailureRecord includes type/schema/key/revision, revision digest, failure code, observed input types, and assembly-policy version; LateEvidenceRecord includes type/schema/episode/latest revision and digest/event, arrival/deadline/reason, raw pointer/root, ordered lineage references, and VersionSet. Host, PID, wall time, and duration are an optional non-canonical envelope. References sort by relation rank, dependency-group bytes, object-before-sentinel, object-ID-or-empty bytes, then digest-or-empty bytes. Optional absence uses a closed sentinel; required absence fails assembly. Assembly input order is revision, feature, logit, support, probability, U, then coverage. Failure precedence is schema, key, duplicate, digest, version, missing, then timeout. Every VersionSet component closes both a version identifier and content digest for schema, parser, formation, detector registry, feature, normalizer, model, calibrator, support, sufficiency, assembly, and manifest. Conforming validation must accept supported, unsupported, late, and assembly-failure records and reject any online record containing `malicious_label`. Online objects reject all label, scenario, marker/window, and ability/operation fields.

Appendix A.2. Episode Algorithm and State Machine

Fixed values are $W_{gap} = 300$ s, $W_{max} = 3600$ s, $L_{wm} = 120$ s, $W_{rev} = 900$ s, and $W_{ret} = 86400$ s. Relations have $\rho = (4, 3, 2, 1)$ for causal/session/endpoint/entity and n_R counts satisfied classes; candidates sort by $(-\rho, -n_R, \Delta_t, t_{start}, id)$. A hub has at least 25 relations in 600 s. Merge permits at most three parents connected by causal, session, or endpoint relations and at most 500 events/200 entities; otherwise attach to the first candidate. The event order is (t^{event}, t^{ingest}, `source_instance`, `event_id`). First finalization freezes t_f and deadline $t_f + 900$ s. The lineage then leaves the open index and remains in a revisable index; at closure, canonical relation keys remain in a closed-lineage index through deadline plus W_{ret}. Each event queries revisable/closed indexes first and returns after exactly one successor/LateEvidenceRecord; only no match permits open start/attach/merge. A qualifying late event before the fixed deadline creates a `finalized` successor without reopening or extending the horizon; the deadline creates a membership-identical `closed` successor. A later related event creates only a LateEvidenceRecord in the late store with reason `beyond_revision_horizon`, `closed_lineage_match`, or `duplicate_after_terminal_outcome`. Replay equality covers every state, late record, output, reference, and digest.

Appendix A.3. Thresholds and Sufficiency Rules

Table A.15 is the single numeric and precedence registry for the submission PDF.

Feature Extractor emits only `detected`, `examined_none`, or `not_examined`. At `EpisodeRevision.event_time_end`, Scope Resolver combines that observation with exactly one content-distinct covering SourceCoverage interval per expected instance and emits `available`, `no_detection`, `collector_unavailable`, `parser_invalid`, or `unknown`; the first two are health-support states, but zero eligible detector values still force out-of-support. CR1 is an overlapping endpoint-to-host binding conflict, CR2 is canonical EDR/NDR flow-direction conflict, and CR3 is conflicting parent/root lineage for one process key. Health is healthy for zero missed 30 s heartbeats, lag ≤ 120 s, valid parser, and live transport; degraded for one missed

Table A.15: Frozen numeric registry and deterministic precedence.

Contract owner	Parameter(s)	Frozen value / order	Validation rule
A. Numeric contract			
Formation	W_{gap}; W_{max}; L_{wm}; W_{rev}; W_{ret}	300; 3600; 120; 900; 86400 s	sealed before final training fit
Formation guards	hub relations/window; parents; component cap	25/600 s; 3; 500 events/200 entities	reject on violation; never learned
Feature	detector IDs; top-k; source mask	3 EDR + 3 NDR; k = 3; EDR+NDR	finite $[0, 1]$; mid-rank CDF
Offline matching	θ_{mal}; θ_{contam}; θ_{match}	.50; .25; .50	freeze before label release
Match predicates	temporal IoU; entity Jaccard	.10; .50	also require exact action/receipt
Uncertainty (paired CI)	bootstrap repetitions; seed	2000; 20260721	resample complete run clusters
B. Deterministic U precedence (first match wins)			
$u_{coverage}$	health/parser state	unknown > unavailable > degraded > complete	evaluate authenticated coverage first
$u_{provenance}$	required references	missing > complete	`partial` reserved; v1 emits only missing/complete
$u_{lateness}$	revision state	late_accepted > revisable > stable	accepted successor; before deadline; closed
$u_{contradiction}$	CR1–CR3	present > unevaluable > none	first applicable conflict rule wins
$u_{support}$	exact support enum	out_of_support; supported_group; supported_global	design-only; no fitted P claimed

heartbeat or lag $(120, 600]$ s; and down for two misses or authenticated stop. Required provenance is every membership/root, selected link, scored input, coverage object, and version; diagnostics are optional. Any later executable threshold file must exactly match Table A.15.

Appendix A.4. Probability Interface Contract (Unevaluated)

Optional score fields may remain in schema. Binding rules only: version identifiers and digests must match; unsupported probability is JSON null; U must not rewrite P; no fitted coefficients are reported in this submission. Affine objectives, group gates, fold assignment, and refit protocols are deferred to supplementary material and are not empirical contributions here.

Appendix A.5. Live-System Lab, Cards, and Offline Labels

The lab, card, and label contract is exactly Sections 4.2–4.3 and Equation 1. It pins public CALDERA and OCSF versions; missing manifest fields fail closed. The amended primary cohort fixes five valid runs per M1–M6 card/mode and five standalone runs per B1–B8 card, all achieved. Each Linux-only M/B card closes image key and digest, ordered ability IDs, no-profile Bash executor plus exact command templates, monotonic offsets, fixed target paths, offline receipt preimage, success/failure predicates, cleanup commands and digest check, and allowed mode. Target commands cannot receive run/card/ability/operation/receipt identifiers or values; orchestrator-only receipts remain in Ground Truth Store. Template variables resolve only from the frozen run manifest. Malicious modes are attack-only or one hash-selected B1–B8 workload begun 60 s earlier; benign cards also run standalone. Missing values and malformed inputs are counted, not zero-filled.

Appendix A.6. Unexecuted Extensions and Statistical Contract

Formation coverage, fragmentation, over-merge, exact-complete recovery, and one-to-one episode F1 are reported for the amended sealed cohort. Fitted calibration, locked-test ranking, RP0–RP7 partial-telemetry children, physical outage, and full assembly cost benchmarks remain registered but unexecuted. The pre-declared group-calibrator gate (50 positive, 50 negative, 10 runs) is not lowered post hoc. If those protocols are later executed, resampling units must be complete runs; undefined and failed outcomes must be counted, never silently zeroed; and protocol amendments must be timestamped before inspecting estimates.

The required supplementary archive is `TrustEpisode_reproducibility_companion_v1.zip`. Concatenating the following two hexadecimal lines gives its SHA-256:

```
ef2081c2a04c3bda0efe518f824ad28f
fb0793a15ca6aeb6d08053b0ec57829b
```

The adjacent sidecar records the same digest.

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